

1. Network Models

The OSI reference model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application. The TCP/IP model has four layers: Link, Internet, Transport and Application.

- The Data Link layer provides framing, error detection (for example CRC) and medium access control.
- The Network layer handles logical addressing and routing (IP).
- The Transport layer provides process-to-process delivery (TCP, UDP), using port numbers to identify applications.

2. Transmission Control Protocol

TCP is a connection-oriented, reliable, byte-stream transport protocol. It provides flow control using a sliding window and retransmits lost segments.

TCP three-way handshake (connection establishment):

1. The client sends a SYN segment with its initial sequence number x .
2. The server replies with SYN-ACK, carrying its own initial sequence number y and acknowledgement number $x+1$.
3. The client sends an ACK with acknowledgement number $y+1$. The connection is now established.

Connection termination uses a four-way exchange: FIN, ACK, FIN, ACK. The side that closes first enters the TIME_WAIT state for twice the maximum segment lifetime (2MSL).

TCP versus UDP: UDP is connectionless and unreliable but has lower overhead and latency, which suits DNS queries, streaming and online games. TCP guarantees ordered, reliable delivery and is used by HTTP, SMTP and FTP.

3. Routing Algorithms

Dijkstra's algorithm finds the shortest paths from a single source to all other nodes in a graph with non-negative edge weights. It is used by link-state routing protocols such as OSPF.

Steps: set the distance of the source to 0 and all others to infinity; repeatedly pick the unvisited node with the smallest tentative distance, mark it visited, and relax all its outgoing edges (if $\text{dist}[u] + w(u,v) < \text{dist}[v]$, update $\text{dist}[v]$). Stop when all nodes are visited.

Time complexity of Dijkstra's algorithm: $O(V^2)$ with a simple array, and $O((V + E) \log V)$ with a binary-heap priority queue. It does not work with negative edge weights.

The Bellman-Ford algorithm handles negative weights in $O(V * E)$ time and is the basis of distance-vector routing protocols such as RIP. Distance-vector routing suffers from the count-to-infinity problem, which is reduced by split horizon and poison reverse.

4. IP Addressing and Subnetting

An IPv4 address is 32 bits long and is written in dotted-decimal notation, for example 192.168.1.10. An IPv6 address is 128 bits long.

CIDR notation writes the prefix length after a slash. A /24 network has $2^8 = 256$ addresses, of which 254 are usable for hosts because the network address and the broadcast address are reserved. The subnet mask of a /24 network is 255.255.255.0.

Number of usable hosts in a subnet = $2^{(32 - \text{prefix})} - 2$.

Private IPv4 ranges are 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. Network Address Translation (NAT) lets many hosts with private addresses share one public address.

5. TCP Congestion Control

TCP congestion control adjusts the congestion window (cwnd) to avoid overloading the network.

- Slow start: cwnd starts at one maximum segment size and doubles every round-trip time (exponential growth) until it reaches the slow-start threshold (sssthresh).
- Congestion avoidance: after sssthresh, cwnd grows by one segment per round-trip time (additive increase).
- On a timeout, sssthresh is set to half of cwnd and cwnd is reset to one segment (multiplicative decrease).
- Fast retransmit: after three duplicate ACKs the sender retransmits the missing segment without waiting for the timeout. With fast recovery (TCP Reno), cwnd is halved instead of being reset to one.

This behaviour is known as AIMD: Additive Increase, Multiplicative Decrease.